RUSTAM KOLUMBAYEV

SOFTWARE ENGINEER — APPLIED AI & LLM SYSTEMS

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Software Engineer with **6+ years** building **AI-powered platforms**, **production LLM systems**, and **scalable backend services** for enterprise clients including Qloo, Pyte, and Jaxel. Active contributor to **frontier-LLM evaluation datasets** via Mercor, authoring codebase-grounded coding tasks and reviewing benchmark artifacts that train and grade frontier coding agents. Expertise in **Python**, **TypeScript**, **LLM integration**, **RAG systems**, **prompt engineering**, **evaluation and benchmark design**, and cloud-native AI/ML infrastructure (AWS, GCP, Azure).

KEY ACHIEVEMENTS

- Cut LLM API costs by **50%** by researching and prototyping **batch prompting architecture** using GPT-4 and Claude, generating natural-language cultural insights at scale while maintaining output quality
- Overhauled AI dashboard frontend end-to-end, slashing load time by **40%** and increasing dashboard adoption by **25%** across enterprise clients consuming ML-driven insights via REST APIs and WebSockets
- Architected UI for **Secure Multi-Party Computation (SMPC)** and enclave-based ML workloads on AWS Nitro and Azure Confidential Containers, enabling Fortune 100 companies to collaborate on AI/ML pipelines with full data privacy
- Drove **SOC 2 Type II** compliance across the AI platform, implementing security controls and audit logging that unlocked **Fortune 100 contracts** in regulated finance and healthcare verticals
- Shipped prompt testing tool (Snodgrass) with Claude and OpenAI integration for AI engineers; application ranked in **top 10% of Y Combinator '24 applicants**
- Author and review **frontier-LLM coding evaluation tasks** and graded rubrics for an AI training data platform (part-time contract)

PROFESSIONAL EXPERIENCE

Qloo, Seattle, WA (Remote) — Feb 2025 — Present

AI cultural intelligence platform using deep learning to analyze 3.7B+ lifestyle entities and trillions of behavioral signals for Fortune 100 clients. "Best Decision Intelligence Company" — 2023 AI Breakthrough Awards.

SENIOR SOFTWARE ENGINEER

Leading frontend and applied AI engineering for Qloo's enterprise platform. Driving LLM integration, real-time ML data visualization, and production system performance.

- **Reduced LLM API costs by 50%** by researching and prototyping batch prompting techniques using GPT-4 and Claude; proposed architecture for processing user queries into contextual, human-readable cultural recommendations at scale
- Produced **interactive ML data visualizations** using React and D3.js, surfacing AI-driven insights via REST APIs and WebSockets; improved insight clarity and boosted dashboard adoption by **25%** across enterprise clients
- **Rebuilt frontend architecture end-to-end**, optimizing rendering strategies and modular patterns; trimmed page load time by **40%** and increased user engagement by **48%** through real-time collaboration features (shared cursors, comments, presence)
- **Led JavaScript-to-TypeScript migration** across core codebase, introducing strict typing and interface-driven development; lowered runtime bugs by **30%** and improved code safety for ML integration layers
- **Spearheaded SOC 2 Type II compliance** implementation, establishing security controls, data handling policies, and audit logging; secured contracts with Fortune 100 clients in finance and healthcare
- Implemented **Sentry** for error tracking and performance monitoring, cutting production issue resolution time by **45%** and proactively catching regressions before user impact

Stack: TypeScript, React, D3.js, Node.js, REST, WebSockets, Sentry

Pyte, Seattle, WA (Remote) — Dec 2022 — Feb 2025

Privacy-preserving AI/ML collaboration platform enabling Fortune 100 companies to run secure computation on sensitive datasets with full compliance.

SOFTWARE ENGINEER

Sole frontend engineer designing and building the UI/UX for a secure ML data collaboration platform (~30 screens). Presented demos that won POC contracts with Fortune 100 companies.

- Crafted cutting-edge UI for **Secure Multi-Party Computation (SMPC)** and enclave-based ML pipelines on **AWS Nitro** and **Azure Confidential Containers**, empowering users to process data in fully encrypted environments in real time
- Deployed real-time secure data collaboration demo with QR code integration; showcased at **Advertising Week NYC (15K+ attendees)** to positive reception, directly improving sales pipeline visibility
- **Decreased deployment bugs by 20%** through automation of end-to-end testing with Cypress, integrated into CI/CD pipelines for continuous quality assurance on secure data-sharing workflows
- Integrated **OAuth (Google, Microsoft)** and secure cloud uploads via AWS S3, GCS, and Azure Blob Storage; resolved penetration-testing vulnerabilities for **SOC 2 compliance**

- Reengineered cross-platform test infrastructure by converting shell scripts to Python, ensuring Mac/Linux compatibility and streamlining CI pipeline execution

Stack: TypeScript, React, Cypress, AWS Nitro, Azure Confidential Containers, OAuth, Python

Jaxel, Seattle, WA (Remote) **May 2021 — Dec 2022**

Enterprise engineering consultancy delivering IT incident management and data analytics platforms for major utility and retail clients.

FULL STACK ENGINEER

- **Boosted data analytics performance by 50%** by building a React-based UI platform for a major electricity provider, optimizing data filtering and eliminating slow database queries; led 2 UI developers across full SDLC
- Achieved **60% reduction in incidents** for a global coffee retailer's inventory management system as head of incident team; oversaw offshore team diagnosing server failures and resolving API issues
- **Cut mean time to resolution (MTTR) by 35%** by instrumenting Datadog dashboards and Postman API monitors across a full-stack React/Node.js application, enabling real-time alerting on 12 critical service endpoints

Stack: React, Node.js, Datadog, Postman

Fixinity, Remote **Jul 2019 — May 2021**

Location-based mobile platform connecting users with local service providers, using real-time data processing and search.

FULL STACK DEVELOPER

- Lowered user complaints by **30%** by implementing **20+ features annually** for real-time search and communication; enhanced satisfaction by **25%** through UX optimization
- Engineered secure authentication system using **Firebase Authentication** and Firestore, onboarding **5,000+ users** in the first 6 months with React, React Native, and Python backend services

Stack: React, React Native, Firebase, Firestore, Python

AI EVALUATION WORK (PART-TIME CONTRACT)

Mercor, Remote **2025 — Present**

AI EVALUATION CONTRIBUTOR

- Write and review **codebase-grounded coding evaluation tasks** and graded rubrics (15–35 atomic criteria each) that frontier AI labs use to train and grade coding agents
- Catch ambiguous prompts, hidden-assumption failures, and grading-leakage risks before tasks ship, improving benchmark reliability

TECHNICAL PROJECTS

Snodgrass — AI Prompt Testing Tool

Prompt evaluation and testing platform for AI engineers with **Claude and OpenAI** API integration. Supports systematic comparison across 50+ prompt variants per session with automated quality scoring and regression detection.

*Impact: Ranked in **top 10%** of Y Combinator '24 applicants.*

ReferrsMe — Referral-Driven Job Platform

Complete frontend for a referral-based job application platform serving **1,000+ active users** with 15+ interactive components in React and TypeScript.

*Impact: Achieved **sub-2-second page loads**.*

Cloud Express — Cloud Deployment UI

Deployment UI that condensed a **14-step manual provisioning workflow into 3 clicks**, for backend infrastructure configuration.

*Impact: Cut average setup time from **45 minutes to under 5 minutes**.*

EDUCATION

KIMEP University — Bachelor of Law — 2019
University of Texas at Austin — Post Graduate Program, Cloud Computing — 2023
Harvard University — CS50x, Introduction to Computer Science — 2019
Nucamp Bootcamp — Full Stack Web and Mobile Development — 2020

TECHNICAL SKILLS

Primary	LLM Integration (OpenAI, Anthropic Claude), Prompt Engineering, RAG Systems, LLM Evaluation & Benchmark Design, Rubric Authoring, RLHF/RLAIF Training Data, Agentic AI Evaluation, Python, TypeScript, React, Node.js, AWS, SOC 2 Compliance
Secondary	JavaScript, Next.js, Express.js, NestJS, FastAPI, Django, D3.js, Redux, Zustand, SQL, PostgreSQL, MongoDB, MySQL, GraphQL, gRPC, WebSockets, TensorFlow.js, Rust, C, WebAssembly, Docker, GCP, Azure (Confidential Containers, Blob Storage), Firebase, Heroku, GitHub Actions, GitLab CI, Cypress, Jest, React Testing Library, Sentry, Datadog, Postman, OAuth 2.0, SMPC, Enclave Computing, Penetration Testing Remediation, mini-swe-agent, SWE-bench style evaluation